Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
12	(a)			**Indicative content** *Information about reaction* • as temperature increases yield decreases • as temperature is increased equilibrium moves to the left / in the endothermic direction • therefore forward reaction is exothermic • as pressure increases yield increases • as pressure is increased equilibrium moves to the right / to the side with fewer gas moles • therefore more gas moles in reactants *Optimum conditions and explanation* • optimum temperature likely to be 100-200°C – higher rate than at lower temperature; sharp fall in conversion above 200°C • optimum pressure likely to be 30-40 atm – no increase in yield above 40 atm • optimum pressure could be 20-30 atm – if costs of plant are significantly greater for higher pressure • compromise required between % conversion and rate i.e. best conversion at around room temperature but rate could be very low • must have information about the reaction rate to determine the optimum conditions • other factors for consideration – catalysts, ease of separation of reactants/products for reactants to be returned to reaction vessel, increased costs of plant to operate at higher pressure	2	2	2	6		